AI-ML DRIVEN MULE ACCOUNT CLASSIFICATION

THE GROWING THREAT OF MULE ACCOUNTS

01 Problems in Banking Fraud Detection

- Traditional monitoring systems rely heavily on fixed, rule-based thresholds.
- Fraud alerts are often triggered only when predefined limits are exceeded.
- Criminals can easily evade detection by splitting transactions into smaller amounts.
- Existing systems struggle to identify complex fraud networks across multiple accounts.

02 Evolving Threats : Rise of Mule Accounts

- Cybercriminals use mule networks to obscure the money trail.
- Funds are moved through multiple accounts within a short time period.
- Organized fraud rings coordinate transactions across different banks and channels.
- Digital banking, instant payments, and online transfers have accelerated the speed of money laundering

03 What Should Be Done? Future Fraud Detection Strategy

- Shift from rule-based systems to AI and Machine Learning-based detection.
- Detect relationships between accounts using network/graph analysis.
- Implement real-time fraud detection and risk scoring.
- Continuously update models to adapt to emerging fraud techniques.

OUR SOLUTION

Hybrid Fraud Intelligence Engine

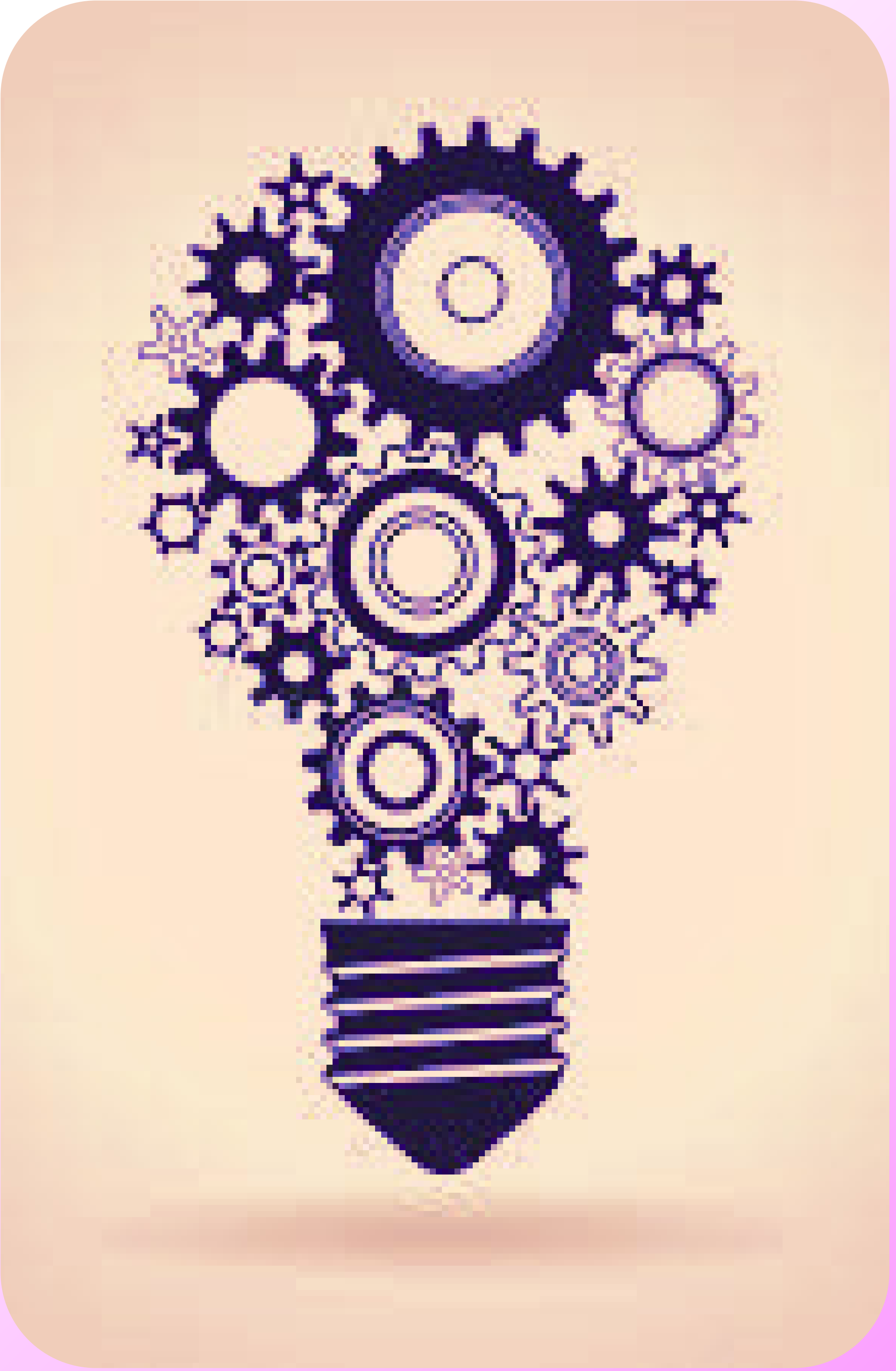
Combines XGBoost, LightGBM, and CatBoost models to identify complex mule-account behavior patterns hidden within 3,923 anonymized banking attributes. The ensemble architecture improves robustness, minimizes false positives, and enhances fraud detection accuracy across diverse transaction profiles.

Advanced Risk Signal Engineering

Generates high-value fraud indicators through statistical feature construction, behavioral pattern analysis, and interaction-based feature discovery. The framework leverages bank-prioritized attributes along with automated feature selection techniques to extract the most predictive risk signals from highly anonymized data.

Explainable Risk Scoring & Compliance Support

Produces a transparent fraud risk score for every account using SHAP-based explainability. Instead of providing only a binary prediction, the system highlights the key behavioral factors responsible for elevated risk, enabling investigators and compliance teams to make informed decisions with full auditability.



SOLUTION FLOW

DATA INGESTION LAYER

- 9,082 customer accounts with 3,923 anonymized banking attributes.
- Captures transaction behavior, account activity, and risk-related patterns.
- Provides labeled data (F3924) for mule account classification.

DATA PROFILING & QUALITY ASESMENT

- Performs data quality validation by identifying missing values, duplicate records, and inconsistencies that may impact model performance.
- Analyzes feature distributions and class balance to ensure reliable risk signal generation and robust fraud detection.

RISK FEATURE CONSTRUCTION ENGINE

- Constructs new risk features using statistical measures and feature interactions.
- Captures behavioral anomalies and transaction patterns associated with suspicious accounts.

HYBRID FRAUD INTELLIGENCE CORE

- Trains an ensemble of XGBoost, LightGBM, and CatBoost to learn complex fraud patterns.
- Uses Stratified K-Fold validation and hyperparameter tuning to maximize detection performance and generalization.

FEATURE DISCOVERY

- Uses XGBoost feature importance and SHAP analysis to identify the most predictive fraud-related attributes.
- Selects high-value signals while removing redundant features, improving model efficiency and detection accuracy.

EXPLAINABLE RISK ANALYTICS

- Generates explainable fraud predictions using SHAP-based feature attribution.
- Highlights the key risk factors contributing to suspicious account classification.

AUTOMATED DECISION

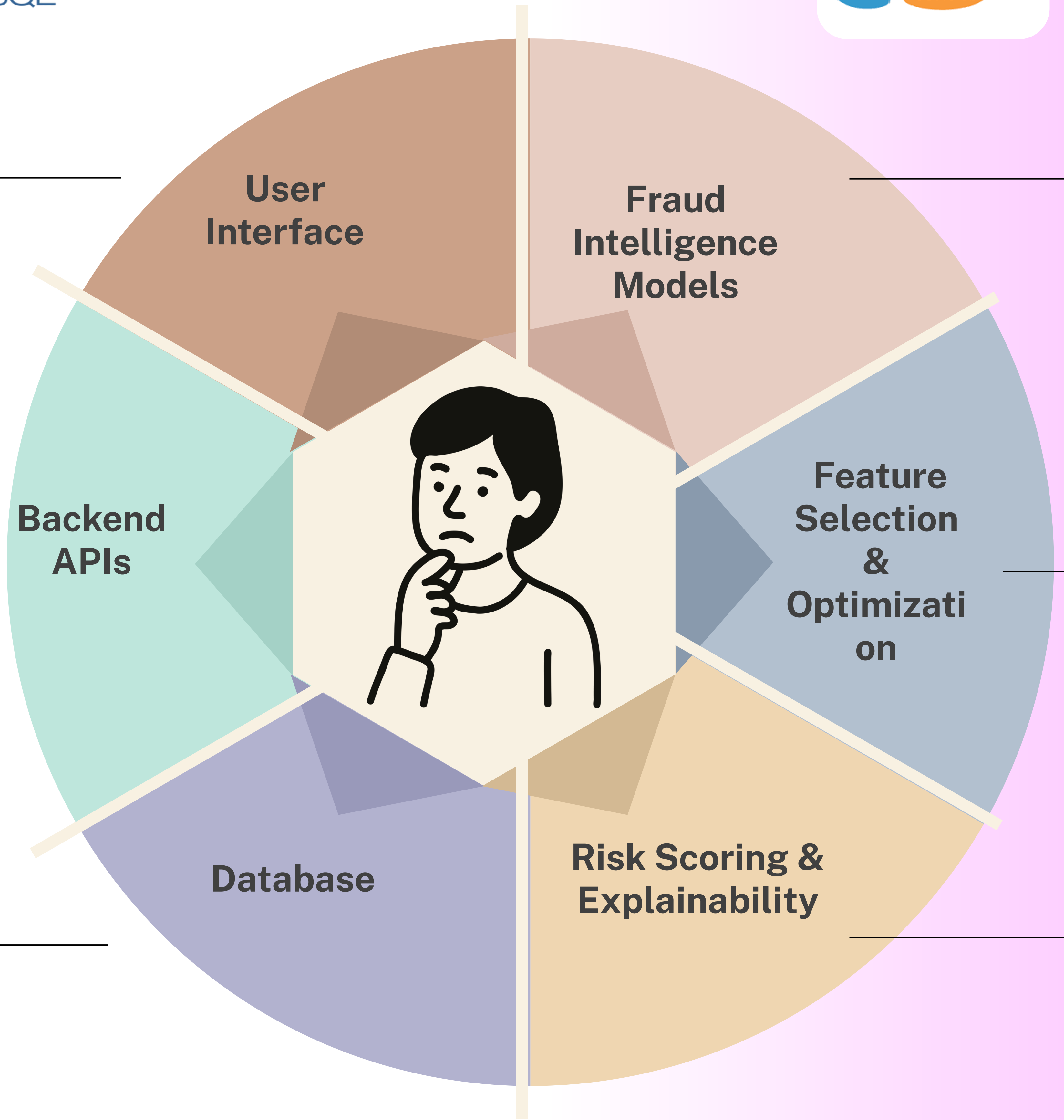
- Translates model predictions into operational risk levels for real-time decision making.
- Supports fraud investigation, compliance monitoring, and proactive mule account prevention.

TECH STACK

ReactJs
CSS
HTML
Vanilla CSS

FastAPI
Python
REST APIs
Docker

PostgreSQL
SQLAlchemy



XGBoost,
LightGBM,
CatBoost Ensemble
for Mule Account
Classification

Scikit-Learn,
SHAP,
Optuna,
tensor flow

SHAP,
Fraud Risk
Scoring,
Decision Support

MEET OUR TEAM!

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